Rahul Krishna Kolayikkath

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ABOUT

I am an AI/ML Engineer with a strong interest in transitioning into academic research on **Natural Language Processing and AI systems**. I enjoy coding, developing projects and conducting experiments. I am motivated by the progress of open-source AI research and aim to actively contribute to the research community. My current research interests lie in designing personalized AI tutors that adapt to individual learners by modeling their understanding, reasoning, and misconceptions.

EDUCATION

2019–2024 **BTech & MTech in Chemical Engineering**
Indian Institute of Technology Madras
CGPA: 8.29
MTech Advisor: Takark Kumar Patra

2021–2022 **Semester Exchange in Computing Science**
University of Groningen
CGPA: 7.70

PROFESSIONAL EXPERIENCE

Tabbie.ai - AI Integration Consultant *(May 2025 - Oct 2025)*
Developed a comprehensive service to evaluate students' answer sheets using foundation models

Flexday.ai - AI Engineer *(Aug 2024 - May 2025)*
- Integrated tree-organized summary retrieval into RAG system enabling it to answer holistic questions
- Implemented contextualized late interactions from BERT to improve retrieval latency by 20 times
- Led a team to develop self-service analytics by building text2SQL and text2Insight solutions

Polymer Research Group, IIT Madras — Undergraduate Researcher *(June 2023 – May 2024)*
- Developed a fully machine-driven Polymer Informatics pipeline inspired by NLP for Master's project
- Trained a BERT-based embedding model on polymer sequences to create molecular fingerprints
- Achieved an accuracy of 0.80 on property prediction by training on 2M polymer sequences

IIT Madras - Teaching Assistant *(Aug 2023 - Dec 2023)*
Taught tutorial classes for the Master's Level Course 'Mathematical Foundations for Data Science'

TCS Research — Research Intern *(June 2022 – July 2022)*
Predicted ingredient–functionality mappings for small organic molecules using deep learning

PUBLICATIONS

Deep Learning Framework for Materials Property Prediction and Design Using QR Codes *(Nov, 2023)*
Developed a QR code based molecular finger print, compared one hot encoding, Morgan finger print and QR-code based finger prints for property prediction. Published in ACS Engineering AU(Co-authored)

PROJECTS

Smart grading *(July, 2024)*
Explored the capabilities of foundation models in identifying misconception and providing feedback

VLMs for image segmentation *(Apr, 2025)*
Built a segmentation pipeline for students' answer segmentation using Molmo(AI2) + Google Gemini

COURSE WORK

Math: Maths Foundations for Data Science, Probability & Statistics, Calculus for Artificial Intelligence
Data Science: Intro to Machine Learning, Artificial Intelligence , Multivariate Data Analysis
Programming: Design and Analysis of Algorithms, Programming & Data Structures in Python

PLATFORMS & PACKAGES

Pytorch, Scikit-learn, TensorFlow, LangGraph, Haystack, NLTK , Git, Docker, AWS, GCP

EXTRA-CURRICULAR ACTIVITIES

Active member of **Institute footbball team**, and NSO (National Sports Organization) Football at IITM